

Operational Boundaries of Aerodynamic Power Rules, Recalibration, and Learned Representations under SCADA Telemetry Degradation

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Abstract—Operating reserve screening near the demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3) is critical for pre-dispatch imbalance risk mitigation in wind-integrated power systems. However, industrial Supervisory Control and Data Acquisition (SCADA) telemetry is vulnerable to transmission latency, packet serialization dropouts, and unobservable pitch registers across commercial aggregator interfaces. While deterministic aerodynamic power-curve quantile rules achieve the lowest reserve-screening surrogate cost under pristine telemetry (589,535 kW · h at immediate dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation), they exhibit systematic reliability breakdown under communication staleness. Under a 60-minute synthetic latency stress test, steep sensitivity changes near rated wind speed amplify state-state errors, surging clean-calibrated physical violation rates to $24.0\% \pm 1.7\%$ ($h = 1$) and $12.20\% \pm 1.25\%$ ($h = 6$). To delineate the operational division of labor between physical rules and machine learning, we establish an empirical operational boundary and value-of-information framework evaluating aerodynamic power rules, state-conditional recalibration, and the Spatio-Temporal Graph Quantile (STGQ) representation family. Disentangling recalibration from neural representations reveals that under full channel observability, simple state-conditional recalibration absorbs 55.3% of shortage loss ($127.6 \rightarrow 57.0$ MWh) at lower cost than neural models. When blade-pitch registers are withheld across aggregator interfaces, learned representations reconstruct operating regimes from electromechanical transients, outperforming recalibrated physics by 46.7k–68.6k kW · h across latencies and doubling boundary recall (0.416 vs. 0.196). Dynamic Mixture-of-Experts routing confers no statistical advantage over unrouted dense baselines ($p = 0.380$), while decoupled modular architectures match tail reliability (9.7% violation). These findings establish that the operational utility of machine learning is localized to partially observable regimes where spatio-temporal representations recover aerodynamic states that simple recalibration cannot resolve.

Index Terms—Wind turbine operating reserve; SCADA telemetry degradation; transmission latency; aerodynamic power curve; reliability breakdown; boundary-risk posterior; selective decision abstention; MPPT-to-pitch transition.

I. INTRODUCTION

In bulk power system operations, real-time generation deficits incur severe asymmetric shortage penalties relative to surplus energy; consequently, symmetric loss metrics like root-mean-square error (RMSE) mask catastrophic tail shortfall events that govern tail shortfall risk and operating reserve adequacy in Level-1 pre-dispatch screening [1], [2], [3], [4], [5]. Reserve screening near the demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-

pitch regulation (Region 3) represents an acute operational bottleneck. Near rated inflow velocity, under the idealized rated-power regulation approximation, local power sensitivity abruptly transitions from a steep cubic trajectory ($\partial P / \partial v \propto v^2$) toward zero as active pitch regulation sheds excess aerodynamic lift [6], [7]. When operating states shift across this threshold unannounced, deterministic power curves over-extrapolate generation, precipitating severe unhedged reserve shortfalls.

In commercial wind plants, Supervisory Control and Data Acquisition (SCADA) telemetry streams face packet serialization overhead, gateway queuing delays, and asynchronous timestamp skews [8], [9]. In the commercial Kelmarsh plant, for instance, 99.6% of turbine status transitions occur strictly between 10-minute SCADA polling boundaries. To stress-test reserve screening resilience, we evaluate controlled counterfactual transmission latency profiles from 10 to 60 minutes: while 10–30 min latencies represent plausible substation buffer backlogs and lossy cellular retransmissions, a 60-min latency is evaluated strictly as a severe contingency outage-envelope stress condition (e.g., gateway failures or cyber-physical storms) to probe the asymptotic breakdown boundary of unadapted physical rules [10], [11]. Furthermore, in third-party aggregator, Virtual Power Plant (VPP), and Transmission System Operator (TSO) pre-dispatch triage, primary blade-pitch registers are frequently restricted across commercial protocol boundaries. Feeding stale or missing sensor measurements into deterministic physical mappings causes systematic misclassification, clearing reserve margins against obsolete operating points.

Existing operational paradigms treat deterministic physical curves and data-driven models as mutually exclusive alternatives, overlooking their complementary operational boundaries under telemetry stress. Under pristine SCADA telemetry, physical quantiles calibrated to turbine aerodynamic specifications achieve the lowest reserve-screening cost among evaluated methods (589,535 kW · h at $h = 1$ and 881,367 kW · h at $h = 6$, with $\sim 6.8\%$ violation). However, deterministic physical rules exhibit systematic reliability breakdown under telemetry staleness: static physical curves over-extrapolate generation near rated wind speed when signals are delayed. Under a 60-min latency stress test, clean-calibrated physical rule violation rates surge to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% target ($q^* = 0.90$)

and leaving 127.6 MWh of unhedged shortfall exposure. Conversely, shallow tabular models capitalize on ultra-short lag autocorrelation at $h = 1$ but undergo severe breakdown across dispatch-grade horizons ($h = 6$) due to inability to model spatial wake advection across turbine arrays [12].

To resolve this operational tension, we establish an empirical operational boundary and value-of-information framework under information freshness constraints. Across 5-seed benchmarks on the 134-turbine WTB commercial plant and cross-farm validations across three European plants, we delineate a clear division of labor: (1) under fresh telemetry, deterministic physical rules deliver the lowest evaluated surrogate cost; (2) under moderate latency with full observability, simple state-conditional recalibration absorbs distribution drift at negligible computational expense; and (3) under unobservable pitch or severe communication degradation, learned spatio-temporal representations reconstruct operating regimes from secondary electromechanical transients, providing essential resilience.

This paper delivers three verifiable contributions: 1. *Quantification of Aerodynamic Rule Reliability Breakdown*: We expose the systematic breakdown of deterministic aerodynamic quantile rules under SCADA transmission latency. Across 5 random seeds on the 134-turbine WTB plant, a 60-min latency contingency surges clean-calibrated physical violation rates from 6.8% to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% target and leaving 127.6 MWh of unhedged shortfall exposure. 2. *Disentangled Value-of-Information Diagnostics* (55.3% vs. 5.4%): Disentangling calibration adaptation from neural representation reveals that state-conditional recalibration absorbs 55.3% of shortage loss ($127.6 \rightarrow 57.0$ MWh) under full observability, making it the most cost-effective recovery. When pitch registers are unobservable across aggregator boundaries, learned representations reconstruct operating states from electromechanical transients, consistently reducing reserve surrogate costs by 46.7k–68.6k kW · h over recalibrated physics and doubling transition recall (0.416 vs. 0.196). 3. *Demystification of MoE Routing and Multi-Farm Envelopes*: We show that dynamic Mixture-of-Experts routing (STGQ-Routed) confers no statistical advantage over unrouted dense baselines (STGQ-Dense, $p = 0.380$), while decoupled modular architectures (STGQ-Modular) match tail reliability (9.7% violation), indicating that operational gains stem from shared spatio-temporal representations and residual quantile calibration rather than dynamic expert gating. Cross-farm evaluations on ENGIE La Haute Borne, Kelmarsh, and Penmanshiel bound reliable deployment to sites with local retraining.

II. RELATED WORK

A. From Average Wind-Power Accuracy to Transition-Window Risk

Spatio-temporal graph neural networks (STGNNs) have advanced multi-step wind plant forecasting by modeling spatial cross-correlations and dynamic wake advection across turbine arrays [12], [13], [14], [15]. Physics-informed extensions integrate engineering wake models and aerodynamic boundary penalties to constrain continuous forecast trajectories [16],

[17], [18]. While these architectures reduce plant-wide L_2 error, continuous trajectory minimization obscures discrete turbine operating state transitions between MPPT (Region 2) and blade-pitch regulation (Region 3) that govern spinning reserve adequacy [19], [20], [21].

Mixture-of-experts (MoE) routing can partition complex feature spaces into localized functional sub-models [22], [23], but unconstrained gating lacks physical interpretability. In power system operations, probabilistic forecasts monetize uncertainty through reserve procurement [1], [2], [24]. This paper addresses an upstream operational bottleneck: screening aerodynamic transition risks under degraded SCADA telemetry prior to market clearing. The resulting reserve evaluation functions strictly as a Level-1 pre-dispatch screening log for balancing-risk triage, abstracting away downstream transmission network power-flow constraints and multi-stage market cashflows.

III. EMPIRICAL OPERATIONAL BOUNDARY FRAMEWORK UNDER INFORMATION FRESHNESS CONSTRAINTS

A. Problem Setup and Mathematical Notation

The formulation decouples multi-step trajectory prediction across future dispatch horizons from anchor-time operating regime identification under degraded telemetry [13], [25]. Let $G = (V, E)$ denote a turbine array spatial graph with N nodes. For each turbine $i \in V$ at discrete time step t , the system observes a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predicts active power $y_{i,t} \in \mathbb{R}$. Given observation history H and forward horizon P , the forecast mapping is:

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}), \quad (1)$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes a time-varying directed graph sequence (WTB) or static graph repeated over time (ERA5). We enforce a strict causal boundary: all input features, graph adjacency structures, and supervisory anchors depend solely on telemetry available at or before dispatch anchor t , with target trajectories commencing strictly at $t+1$. In WTB, instantaneous active power $\text{Patv}_{i,t}$ serves solely as an anchor-time operational status indicator, preventing look-ahead leakage. Routing operates at node level: each turbine i at dispatch anchor t receives an independent routing distribution $\mathbf{g}_{i,t}$, accommodating heterogeneous aerodynamic states across perimeter and wake-shaded interior turbines.

B. Operating-Boundary Physical Anchors and Power-Curve Baseline

Supervisory anchors are grounded where aerodynamic governing equations exhibit distinct, well-posed regimes, whereas ambiguous or transitional states remain guided by prediction loss and spatial graph regularization [8], [26], [27].

Aerodynamic power conversion and control cliff: In variable-speed wind turbine aerodynamics, mechanical power captured from inflow air is governed by $P_{\text{mech}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), \beta(t)) v^3(t)$, where $\rho_{\text{air}} \approx 1.225 \text{ kg/m}^3$, R is rotor radius, $v(t)$ is wind speed, $\beta(t)$ is blade pitch angle, and $\lambda(t) = \omega_r(t)R/v(t)$ is tip-speed ratio. Standard aeroelastic turbine models approximate $C_p(\lambda, \beta)$ via non-linear empirical polynomials [6], [7] (formalized in Supplementary Appendix A). Turbine dynamics are partitioned by rated wind speed u_{rated} : in Region 2 (MPPT, $u_{\text{idle}} \leq v \leq u_{\text{rated}}$), blades

maintain optimal fine pitch $\beta \approx 0^\circ$ yielding cubic power growth $P(v) \propto v^3$; in Region 3 (Pitch regulation, $u_{\text{rated}} < v \leq u_{\text{cut-out}}$), the pitch controller rotates blades to shed lift, clamping electrical output at nameplate capacity P_{rated} . In idealized aerodynamic representations, this transition exhibits a localized slope discontinuity [6]:

$$\left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^-} = \frac{3}{2} \rho_{\text{air}} \pi R^2 C_{p,\text{max}} u_{\text{rated}}^2 \gg 0, \quad \left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^+} \approx 0. \quad (2)$$

When SCADA telemetry experiences latency $\tau > 0$, deterministic curves evaluate stale state tuples $(v_{t-\tau}, \beta_{t-\tau})$. When inflow enters Region 3 while delayed pitch readings remain at $\beta = 0^\circ$, this sensitivity cliff amplifies stale estimation errors: the deterministic curve over-extrapolates generation as $\hat{P}_{\text{phys}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), 0^\circ) v^3(t) \gg P_{\text{rated}}$, driving shortage violation rates to $24.0\% \pm 1.7\%$ under Delay-6.

Aerodynamic operating regimes and turbine-calibrated anchors: Let $\bar{p}_{i,t} = \frac{1}{3} \sum_{k=1}^3 p_{i,t}^{(k)}$ denote average blade pitch angle across all three blades for turbine i at dispatch anchor t , and $w_{i,t} = \text{Wspd}_{i,t}$ anemometer wind speed. The declared operating boundary is defined with turbine-calibrated cut-in threshold u_{idle} , rated wind speed u_{rated} , and pitch-activation angle p_{th} :

$$R_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transitional/ambiguous}). \end{cases} \quad (3)$$

Thresholds are calibrated to turbine specifications: WTB (134 turbines: $u_{\text{idle}} = 3.0$ m/s, $u_{\text{rated}} = 10.5$ m/s, $p_{\text{th}} = 2.0^\circ$), Kelmarsh (6 MM92 turbines: 3.0 m/s, 12.5 m/s, 1.0°), Penman-shiel (15 MM82 turbines: 3.0 m/s, 14.5 m/s, 1.0°), and ENGIE La Haute Borne (4 MM82 turbines: 3.0 m/s, 14.5 m/s, 1.0°). Only well-defined regimes ($R_{i,t} \in \{0, 1, 2\}$) provide direct supervisory alignment; transitional samples are masked via $M_{i,t} = \mathbf{1}[R_{i,t} \neq 3]$. These define our **Aerodynamic Power-Curve Baseline**. Blade pitch $\bar{p}_{i,t}$ and wind speed $w_{i,t}$ serve strictly as *offline supervisory anchors* to delineate ground-truth regimes and calibrate baseline curves; online inference relies strictly on available telemetry $\mathbf{X}_{t-H+1:t}$. In pitch-withheld evaluations (Section V-E), $\bar{p}_{i,t}$ is strictly excluded from $\mathbf{X}$, and the network reconstructs regimes from secondary electromechanical channels (active/reactive power) and dynamic wake topology (recovering NMI = 0.561).

State-conditional quantile recalibration: To evaluate whether lightweight non-neural adaptation resolves latency without deep representation learning, we formalize state-conditional recalibration. Given delayed telemetry $\mathbf{s}_{i,t-\tau} = (w_{i,t-\tau}, \bar{p}_{i,t-\tau})$, we partition the state space into K operational bins $\{\mathcal{B}_k\}_{k=1}^K$ defined by cut-in, rated wind speed, and pitch thresholds ($R_{i,t-\tau} = k$). To eliminate data leakage, empirical quantile reserve margins $\hat{r}(k)$ are calibrated strictly on held-out validation residuals $\mathcal{D}_k^{\text{val}}$:

$$\hat{r}(k) = \inf \left\{ r \in \mathbb{R} : \frac{1}{|\mathcal{D}_k^{\text{val}}|} \sum_{(i,t) \in \mathcal{D}_k^{\text{val}}} \mathbf{1}[y_{i,t}^{\text{phys}} - y_{i,t} \leq r] \geq q^* \right\}, \quad (4)$$

where $\mathcal{D}_k^{\text{val}} = \{(i,t) \in \mathcal{D}_{\text{val}} : \mathbf{s}_{i,t-\tau} \in \mathcal{B}_k\}$ and $q^* = 1 - 1/\rho = 0.90$. We formalize two distinct calibration protocols: (1) *Clean-Frozen Calibration*, where reserve margins $\hat{r}^{\text{clean}}(k)$ are calibrated exclusively on nominal clean validation data ($\tau = 0$) and strictly frozen prior to deployment, quantifying unadapted vulnerability under latency ($\tau > 0$); and (2) *Condition-Matched Validation Calibration*, where for a known delay τ , reserve margins $\hat{r}(k; \tau)$ are re-estimated on held-out validation residuals undergoing the identical delay τ , establishing a non-neural benchmark that absorbs distribution drift under full channel observability. Condition-matched validation calibration assumes that the degradation class is identified and a condition-matched validation calibration set is available; it is therefore evaluated as an adaptation benchmark rather than a zero-adaptation robustness setting. Observing delayed telemetry in bin $\mathcal{B}_k$ yields reserve allocation $\hat{r}_{i,t} = \hat{r}(k)$. ERA5 is retained as an observability contrast where sensible heat flux directly marks convective regimes (Supplementary Material A).

C. Spatio-Temporal Graph Quantile (STGQ) Framework and Boundary Representation

Physical dynamic wake graph: When primary SCADA telemetry degrades, spatial coupling across turbine arrays provides physical redundancy to infer local operating regimes [13], [25]. For WTB, we construct a time-varying directed wake graph $\mathcal{A}_t$; candidate turbine pairs (i,j) are filtered by physical distance ($d_{ij} \leq d_{\text{max}}$), activated when upstream turbine j lies within an aerodynamic wake cone aligned with instantaneous local wind direction θ_t (half-angle $\alpha = 25^\circ$), weighted by streamwise decay $\exp(-d_{\parallel}/\sigma_x)$ and cross-stream decay $\exp(-d_{\perp}^2/2\sigma_y^2)$, and pruned to the K strongest inbound wake neighbors [16], [18]. This defines an instantaneous aerodynamic wake score $s_{i,t}^{\text{wake}}$, providing spatial context even during anemometer corruption.

Directed-diffusion GRU encoder: With 110k parameters and an average single-window execution time of 3.8 ms in desktop benchmarking, the encoder features a compact parameter footprint that may align with local substation automation computing envelopes, though physical deployment on embedded RTU hardware remains unvalidated. Two directed diffusion blocks aggregate self, inbound, and outbound messages across each graph snapshot:

$$\mathbf{X}_t^{(\ell+1)} = \text{LN} \left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)} \right), \quad (5)$$

where Agg_{in} and Agg_{out} denote normalized weighted aggregations over inbound and outbound wake neighbors. A GRU then extracts node-level representations $\mathbf{h}_{i,t}$.

We formalize the Spatio-Temporal Graph Quantile (STGQ) framework as a unified representation family comprising three architectural configurations: (1) **STGQ-Dense**, an unrouted baseline coupling spatial graph embeddings $\mathbf{h}_{i,t}$ to a shared linear projection for point forecasts $\hat{\mathbf{y}}_{i,t+1:t+P}$ and a joint dense quantile head; (2) **STGQ-Routed**, an architectural comparator evaluating dynamic Mixture-of-Experts (MoE) routing, where gating logits $\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}])$ compute soft routing weights $\mathbf{g}_{i,t} = \text{softmax}(\mathbf{z}_{i,t}/\tau_{\text{gate}})$ to combine specialized

expert outputs $\hat{y}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t})$; and (3) **STGQ-Modular**, a decoupled architecture pairing frozen spatio-temporal backbone representations with a dedicated residual quantile head estimating reserve margin $\hat{r}_{i,t}$ via pinball loss ($q^* = 0.90$). This taxonomy isolates whether diagnostic reserve resilience stems from dynamic expert routing, shared spatio-temporal representations, or decoupled residual calibration.

Cross-sensor electromechanical diagnostic mechanism:

When blade-pitch telemetry $\bar{p}_{i,t}$ suffers packet loss, transmission latency, or protocol unobservability across aggregator boundaries, the dynamic graph encoder leverages active power transients (Patv), terminal voltage fluctuations, reactive dynamics, and upstream wake propagation to reconstruct the probability of active blade-pitch regulation. In nominal operation, the boundary-risk head ingests $\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}]$. Under telemetry impairment, the learned representation provides structural defense-in-depth, mitigating the severe shortage volumes that compromise deterministic rules.

D. Integration of Physical Constraints into Model Optimization

Prediction loss alone cannot identify regime-specialized representations because unconstrained models fit averaged dynamics without physical semantics [22]. We employ a structured multi-task objective combining multi-step trajectory prediction loss $\mathcal{L}_{\text{pred}}$ with auxiliary physical regularizers: $\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}$ (formalized in Supplementary Appendix A). Here, $\mathcal{L}_{\text{align}}$ anchors routing logits to declared regimes $R_{i,t}$ on unmasked samples using train-only inverse-frequency weights; $\mathcal{L}_{\text{force}}$ sharpens discrimination between MPPT and pitch regulation; $\mathcal{L}_{\text{bal}}$ prevents expert starvation; and $\mathcal{L}_{\text{aux}}$ with $\mathcal{L}_{\text{smooth}}$ enforce spatial wake consistency. The normalized issue-time boundary-risk posterior is computed from gate probabilities across operational regimes as $\pi_{i,t}^{\text{pitch}} = g_{i,t}^{(\text{pitch})} / \sum_{c \in \mathcal{C}_{\text{op}}} g_{i,t}^{(c)}$. Checkpoint selection is strictly governed by validation RMSE, preventing auxiliary penalties from distorting prediction accuracy.

E. Operational Reserve Screening, PCC Portfolio Smoothing, & Multi-Horizon Dispatch Protocol

Boundary-aware routing supports pre-dispatch decisions by screening imbalance risks and sizing upward spinning reserves to prevent costly shortage penalties [24], [28].

Two-level hierarchical grid dispatch interface: To formalize the operational boundary of this work, we distinguish a two-level grid dispatch hierarchy: *Level 1* (Local Pre-Dispatch Risk Filtering & Screening Layer, Paper Scope), operating at the wind plant EMS or aggregator terminal to ingest SCADA telemetry, forecast spatio-temporal power, diagnose operational regime boundaries under telemetry degradation, and size pre-dispatch reserve allocations mitigating unhedged balancing exposure; and *Level 2* (Grid-Level Physical Clearing & Network-Constrained Dispatch, Downstream Interface), managed by the TSO to execute network-constrained economic dispatch (NCED), unit commitment, and full AC-OPF subject to bus voltage security, transmission thermal limits, and bulk

reserve margins. Our formulation is strictly bounded to Level 1 pre-dispatch risk screening proxies, abstracting away transmission network power-flow constraints and market cashflow settlements.

Penalized Reserve-Shortfall Energy metric and Newsvendor derivation: For scheduled point forecast $\hat{y}_{i,t}$ and realized generation $y_{i,t}$, over-forecast shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. Sizing upward reserve margin r_b incurs a reserve procurement energy equivalent $r_b \Delta t$ and an asymmetric shortage penalty $\rho \max(s_{i,t} - r_b, 0) \Delta t$. For penalty ratio ρ , the total operational proxy metric is formalized as the **Penalized Reserve-Shortfall Energy Index (PSREI)** in kW · h (evaluated at representative proxy ratio $\rho = 10$, alongside sensitivity evaluations across $\rho \in \{5, 10, 20\}$):

$$C(r_b; \rho) = \sum_{(i,t)} [r_b(i, t) + \rho \max(\hat{y}_{i,t} - y_{i,t} - r_b(i, t), 0)] \Delta t. \quad (6)$$

PSREI is a decision-oriented risk surrogate targeting pre-dispatch shortage mitigation rather than financial settlement. Differentiating $\mathbb{E}[C(r_b; \rho)]$ with respect to reserve margin r_b yields the first-order optimality condition $\partial \mathbb{E}[C] / \partial r_b = 1 - \rho \Pr[s_{i,t} > r_b] = 0$, giving the classical Newsvendor critical fractile $q^*(\rho) = 1 - 1/\rho$ [1]. For our representative proxy ratio $\rho = 10$, $q^*(10) = 0.90$, establishing a nominal 10% violation target.

Loss geometry and pinball quantile head: Conventional point forecasting optimizes symmetric quadratic loss $\mathcal{L}_{L_2}(y, \hat{y}) = (y - \hat{y})^2$, whose Bayes-optimal predictor converges to conditional expectation $\mathbb{E}[y | \mathbf{x}]$. In bounded wind power regimes with non-Gaussian transition densities, the conditional mean systematically departs from upper conditional quantiles, while balancing operations ($\rho = 10$) penalize generation shortfalls ten times more heavily than surplus headroom. Upward reserve margin $\hat{r}_{i,t}$ required to cover shortfall residual $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$ is estimated via asymmetric pinball loss:

$$\mathcal{L}_q(s, \hat{r}) = (s - \hat{r}) (q - \mathbf{1}[s < \hat{r}]) = \begin{cases} q(s - \hat{r}), & s \geq \hat{r}, \\ (1 - q)(\hat{r} - s), & s < \hat{r}. \end{cases} \quad (7)$$

Setting $q = q^* = 0.90$ aligns training with the Newsvendor condition: Bayes-optimal prediction converges to conditional shortfall quantile $\hat{r}^* = Q_{0.90}(s | \mathbf{x})$, targeting a nominal 10% exceedance probability [1], [2].

PCC portfolio smoothing & multi-horizon dispatch: Power delivered to the bulk grid is metered at the Point of Common Coupling (PCC) bus: $P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$ and $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$. Aggregating decentralized turbine power flows at the PCC bus unlocks spatial portfolio diversification: localized high-frequency aerodynamic turbulence cancels out across the array, leaving boundary regime shifts as the primary source of unhedged reserve risk. We evaluate two distinct operational horizons: (1) **Immediate dispatch** ($h = 1$, 10-min ahead), representing real-time economic dispatch (RTED) dominated by autocorrelation; and (2) **Operational dispatch** ($h = 6$, 1-hour ahead), representing intra-day clearing and

storage scheduling where spatial wake dynamics and weather transitions are paramount.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and Preprocessing

Empirical evaluation spans two contrasting observability settings: the 134-turbine WTB commercial wind plant benchmark and the ERA5 thermodynamic contrast patch. WTB (KDD Cup 2022) comprises 134 wind turbines and 245 days of continuous 10-minute SCADA telemetry ($T = 35,280$, $N = 134$, $F = 11$) [29], where turbulent inflow, wake propagation, and turbine pitch regulation interact nonlinearly, rendering the MPPT-to-pitch transition indirectly observable. Missing inputs are forward-filled per turbine and mean-imputed; non-positive power values are masked out of supervision. ERA5 covers three archived months on a 16×16 spatial grid ($T = 2208$, $N = 256$, eight features) [30], where surface sensible heat flux directly marks convective regimes. Both datasets adopt an observation history of $H = 36$ and a prediction horizon of $P = 24$ under chronological train/validation/test splits (180/30/35 days for WTB; 1325/441/442 frames for ERA5).

B. Techno-Economic Validation and Benchmarking Framework

The empirical benchmarking protocol evaluates two complementary axes: (1) *Isolating routing mechanisms under matched parameter capacity*: Comparing unrouted dense representations (STGQ-Dense), dynamic Mixture-of-Experts routing (STGQ-Routed), and decoupled modular architectures (STGQ-Modular) to determine whether dynamic expert routing adds operational value beyond shared spatio-temporal representations; and (2) *Benchmarking forecasting accuracy and reserve cost against established spatio-temporal baselines*: Evaluating Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE [31], [32], [33], alongside persistence, deterministic aerodynamic curves, and GBDT lag models (detailed in Supplementary Appendix A).

C. Training Protocol and Evaluation Metrics

All neural architectures share an identical training protocol using AdamW with early stopping on validation RMSE, gradient clipping, and mixed-precision execution on a single GPU across five strict-mask random seeds (201–205). Strong baselines (Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, TiDE) repeat across seeds 201–205 (30 completed runs under identical evaluation caches). To eliminate data leakage, the experimental pipeline strictly enforces a sequential three-split dataflow: $\mathcal{D}_{\text{train}} \xrightarrow{\text{train } \theta} \mathcal{D}_{\text{val}} \xrightarrow{\text{calibrate } \hat{r}(k)} \text{Freeze } \{\theta, \hat{r}\} \xrightarrow{\text{causal replay}} \mathcal{D}_{\text{test}}$. Quantile offsets $\hat{r}(k)$ are estimated strictly on held-out validation residuals $\mathcal{D}_{\text{val}}$ and frozen prior to test replay.

We establish a dual-level resampling hierarchy: (1) *Model Initialization Robustness*, reporting mean $\pm$ standard deviation across five random seeds (201–205); and (2) *Operational Time-Series Uncertainty*, evaluating operational risk metrics via a day-block bootstrap over the 35 test days (1,000 resamples of 144-step daily blocks, preserving diurnal cycles and serial autocorrelation) to construct empirical 95% confidence intervals and paired difference distributions. Evaluation spans:

point accuracy (MAE/RMSE, switch-window RMSE within ± 1.0 m/s of rated wind speed); physical regime alignment (NMI, ARI, gate entropy); and operational reserve consequence (PSREI reserve-shortfall cost, empirical violation rate, unhedged shortage MWh).

D. Controlled Telemetry Degradation Protocols and Realism Mapping

To stress-test reserve screening resilience under adverse communication conditions, we evaluate controlled counterfactual telemetry degradation protocols alongside sensor perturbation controls across all five seeds (201–205): (1) **Tier 1 (Nominal SCADA Polling, 0–10 min, Clean)**: fresh telemetry on the 10-min SCADA grid (IEC 61400-25); (2) **Tier 2 (Plausible Degraded Telemetry, 10–30 min, Delay-1 to Delay-3)**: communication latencies induced by substation ring-buffer backlogs and lossy cellular retransmissions [9]; and (3) **Tier 3 (Contingency Stress-Test Upper Envelope, 60 min, Delay-6)**: a 6-step (60 min) synthetic transmission lag evaluated strictly as an upper contingency stress envelope (simulating gateway failures or cyber-physical storms) [10], [11] to establish the asymptotic breakdown boundary of unadapted physical rules. Sensor perturbation controls comprise additive Gaussian noise ($\sigma_v = 1.0$ m/s, $\sigma_\beta = 2.0^\circ$) and a two-state Markov-Gilbert burst dropout chain ($p_{GB} = 0.08$, $p_{BB} = 0.75$, $d \leq 6$) simulating transient wireless dropouts [9], [11].

Fig. 1 illustrates the operational dispatch setting: graph topology reveals how prediction errors propagate across turbine rows, while regime-anchor planes define the physical boundaries governing reserve screening.

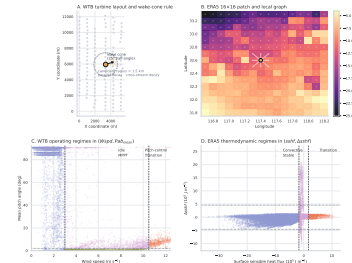


Fig. 1. Operating-decision context and physical regime anchors: (A) WTB layout with wake cone geometry (25° half-angle); (B) ERA5 patch with sensible heat flux; (C) WTB operating regimes in $(Wspd, Pab_{\text{mean}})$ plane with MPPT-to-pitch boundary; (D) ERA5 thermodynamic regimes in $(sshf, \Delta sshf)$ plane.

V. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The empirical evaluation contrasts high-fidelity physical aerodynamic priors against deep spatio-temporal graph fallbacks across multi-horizon dispatch environments. All models are evaluated across five declared random seeds (201–205) under strict automated verification gates (zero directory leakage, full telemetry completeness, cardinality $N = 5$, exact checksums).

A. Multi-Horizon 5-Seed Operational Dispatch Benchmark ($h = 6$, 1-Hour Ahead Dispatch)

Table I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) across 134 turbines over the 35-day test split. Table II reports seed-paired cost differences $\Delta \text{Cost} = \text{Cost}_{\text{Baseline}} - \text{Cost}_{\text{STGQ-Routed}}$, where positive values denote surrogate reserve penalty reductions

delivered by STGQ-Routed, along with 95% bootstrap confidence intervals.

1) Comparative Architecture Analysis: Dynamic Routing versus Dense Representations

Benchmarking capacity-matched neural architectures directly tests whether dynamic expert routing provides operational benefits over unrouted dense heads. Under nominal Clean telemetry, STGQ-Routed ($959,027 \pm 103,290$ kW · h at $h = 6$; $660,990 \pm 80,749$ kW · h at $h = 1$) and STGQ-Dense ($991,181 \pm 110,211$ kW · h at $h = 6$; $690,614 \pm 71,031$ kW · h at $h = 1$) achieve statistical parity. At $h = 6$, the seed-paired cost difference is $+32,154$ kW · h with a bootstrap 95% confidence interval crossing zero ($[-1,148, +65,456]$ kW · h, Table II). At $h = 1$, the Dense/MoE cost ratio is 1.045, and the seed-paired bootstrap interval ($+29,624$ kW · h) likewise crosses zero ($[-53,774, +113,023]$ kW · h, $p = 0.380$), matching the Markov burst regime ($p = 0.237$). Under persistent 6-step latency (Delay-6, $h = 1$), STGQ-Routed shows a modest cost reduction over STGQ-Dense ($1,412,321$ vs. $1,491,958$ kW · h; cost ratio 1.056, paired difference $+79,637$ kW · h, uncorrected $p = 0.038$, non-significant under Bonferroni $\alpha = 0.0125$) while maintaining virtually identical empirical violation rates ($10.6\% \pm 1.1\%$ for STGQ-Routed vs. $10.5\% \pm 0.5\%$ for STGQ-Dense). Decoupled modular architectures (Section V-A3) achieve $9.7\% \pm 1.1\%$ violation in Delay-6 without expert routing, confirming that diagnostic reserve utility arises from shared spatio-temporal representations and residual calibration rather than dynamic gate routing.

2) Reliability Breakdown of Deterministic Physical Rules under Stale Telemetry

Under clean telemetry, Continuous Physical Quantile achieves the lowest baseline PSREI cost ($589,535 \pm 74,886$ kW · h at $h = 1$ and $881,367 \pm 95,741$ kW · h at $h = 6$) with well-calibrated tail coverage ($6.81\% \pm 1.13\%$ violation at $h = 6$). However, Table I exposes a **systematic reliability breakdown** under telemetry staleness. Under 6-step latency (Delay-6), clean-calibrated physical quantile rules suffer a violation explosion to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% violation target ($q^* = 0.90$). This inflates operational reserve costs to $1,136,466 \pm 84,101$ kW · h at $h = 6$ (incurring $47,413 \pm 3,091$ kW · h of unhedged shortage) and $1,656,285$ kW · h at $h = 1$ (leaving 127.6 MWh of unhedged shortfall exposure). Adapting validation calibration to degraded operating states alone cuts shortage from 127.6 MWh (24.0% violation) to 57.0 MWh (10.7% violation), absorbing 55.3% of the shortfall exposure and recovering physical reserve cost to $1,228,609 \pm 71,940$ kW · h (9.46% fleet violation). STGQ-Routed further curtails shortage to 53.9 MWh ($10.6\% \pm 1.1\%$ violation), delivering an incremental 5.4% shortage reduction under matched calibration (combined 57.7% reduction vs. clean-calibrated physics) while saving 190,590 kW · h in total cost.

3) Decoupled Modular Architectures and Tail Conservatism Analysis

In contrast to end-to-end task optimization, Frozen-Embedding Direct Quantile MLP illustrates the pathology of naive decoupled quantile regression: direct pinball regression on frozen spatial embeddings without calibration incurs massive reserve over-estimation ($1,477,967$ to $1,511,794$ kW · h, a 44.57% to

72.02% penalty over STGQ-Routed, Table II). Calibrated modular architectures (Supplementary Table A11d) resolve this tail conservatism: Cascaded Frozen MLP achieves $15.528\text{M} \pm 1.599\text{M}$ kW · h across the multi-horizon 24-step dispatch envelope (vs. 16.065M kW · h for STGQ-Routed; paired difference -0.537M , 95% CI $[-1.988\text{M}, +0.915\text{M}]$ crosses zero); Independent Consequence MLP yields $15.805\text{M} \pm 1.680\text{M}$ kW · h; and the true STGQ-Modular architecture (Decoupled Residual Quantile Head pairing the frozen backbone with calibrated residual quantile regression) achieves $1,284,098 \pm 114,629$ kW · h (1.284M) and $9.7\% \pm 1.1\%$ violation in Delay-6 ($h = 1$, Table III), matching or surpassing end-to-end STGQ-Routed ($1,412,321$ kW · h, $10.6\% \pm 1.1\%$). Hence, diagnostic reserve benefits do not require dynamic MoE routing: modular architectures with calibrated consequence mappings deliver equivalent tail reliability without dynamic gating overhead.

B. Horizon Boundary Diagnosis ($h = 1$, 10-Minute Immediate Dispatch vs. $h = 6$)

Comparing immediate 10-minute dispatch ($h = 1$) against 1-hour ahead dispatch ($h = 6$) exposes a fundamental horizon disconnect across model families (Supplementary Tables A11j/A11k). At $h = 1$, shallow decision trees (Missingness-Aware GBDT) capitalize on ultra-short wind-speed autocorrelation, achieving the lowest data-driven cost ($574,822 \pm 79,685$ kW · h in Clean with 7.6% violation). Across dispatch-grade horizons ($h = 6$), however, GBDT undergoes severe breakdown: operational costs inflate to $1,238,603$ kW · h in Clean and $1,320,003$ kW · h in Delay-6, trailing STGQ-Routed by $+156,411$ to $+287,563$ kW · h across all regimes (95% block bootstrap CIs strictly excluding zero, Table II) due to inability to model spatial wake advection. At $h = 1$, STGQ-Dense and STGQ-Routed exhibit statistical parity in Clean ($+29,624$ kW · h, $p = 0.380$), with a modest margin in Delay-6 ($+79,637$ kW · h, non-significant under Bonferroni correction). Across both horizons, unadapted physical rules collapse under Delay-6 ($24.0\% \pm 1.7\%$ at $h = 1$, $12.20\% \pm 1.25\%$ at $h = 6$), requiring dynamic state adaptation.

C. Multidimensional Operational Boundaries and Selective Decision Abstention

Scanning transmission delays $\tau \in \{0, 10, 20, 30, 60\}$ min, horizons $h \in \{1, 3, 6\}$, and pitch observabilities (a11 vs. no_pitch) across five seeds on WTB delineates a **Three-Regime Operational Framework** (Table III): 1. *Regime I (Simple Recalibration Sufficient under Full Observability)*: Under fresh telemetry ($\tau = 0$, a11), continuous physical quantiles achieve the lowest reserve cost ($593,258 \pm 74,886$ kW · h at $h = 1$, within 0.6% of the primary $589,535$ kW · h benchmark, with $7.29\% \pm 1.42\%$ violation), whereas neural representations incur an 11.8% cost penalty ($663,347$ kW · h) without reliability gain. When latency increases to $\tau \in [10, 30]$ min, clean physical rules collapse (11.2% – 16.6% violation), but state-conditional recalibration fully restores physical compliance (7.1% – 7.5%) at lower cost than neural models ($742,211$ – $939,190$ kW · h vs. $786,366$ – $987,760$ kW · h). 2. *Regime II (Learned Representation Advantage under Unobservable Pitch)*: When blade-pitch telemetry is withheld across aggregator or OEM boundaries (no_pitch), physical aerodynamic rules lose direct rotor

TABLE I

CROSS-SEED MULTI-REGIME OPERATIONAL DISPATCH BENCHMARK UNDER CLEAN-CALIBRATED PROTOCOL ($h = 6$, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN $\pm$ STANDARD DEVIATION ACROSS FULL TEST SPLIT (35 DAYS, 134 TURBINES, $\Delta t = 10$ MIN, COST RATIO $\rho = 10$). ALL QUANTILE BINS AND MULTIPLIERS CALIBRATED EXCLUSIVELY ON NOMINAL CLEAN VALIDATION DATA.

Regime	Evaluated Model	Penalized Reserve-Shortfall Energy (kW-h, Proxy at $\rho = 10$)	Violation Rate	Reserve (kW)	Shortage (kW-h)	Pinball Loss
Clean	Continuous Physical Quantile	881,367 $\pm$ 95,741	6.81% $\pm$ 1.13%	653,738 $\pm$ 72,557	22,763 $\pm$ 4,013	31.41 $\pm$ 2.66
	Frozen-Embedding Direct Quantile MLP [‡]	1,649,721 $\pm$ 285,682	0.58% $\pm$ 0.26%	1,630,572 $\pm$ 293,346	1,915 $\pm$ 958	64.62 $\pm$ 12.05
	Global Quantile	998,988 $\pm$ 107,360	6.46% $\pm$ 1.31%	771,304 $\pm$ 95,661	22,768 $\pm$ 4,854	36.50 $\pm$ 3.41
	STGQ-Dense	991,181 $\pm$ 110,211	6.54% $\pm$ 1.04%	761,480 $\pm$ 94,244	22,970 $\pm$ 4,121	36.16 $\pm$ 3.58
	STGQ-Routed	959,027 $\pm$ 103,290	7.10% $\pm$ 1.17%	707,249 $\pm$ 91,305	25,178 $\pm$ 4,816	34.77 $\pm$ 3.16
	Missingness-Aware GBDT	1,238,603 $\pm$ 112,051	4.50% $\pm$ 0.73%	1,085,168 $\pm$ 118,015	15,343 $\pm$ 2870	46.85 $\pm$ 3.48
Delay-6	Continuous Physical Quantile	1,136,466 $\pm$ 84,101	12.20% $\pm$ 1.25% [†]	662,335 $\pm$ 73,483	47,413 $\pm$ 3,091	38.85 $\pm$ 2.06
	Frozen-Embedding Direct Quantile MLP [‡]	1,682,251 $\pm$ 284,323	1.20% $\pm$ 0.52%	1,651,979 $\pm$ 297,715	3,027 $\pm$ 1,594	62.15 $\pm$ 11.70
	Global Quantile	1,180,603 $\pm$ 91,754	10.85% $\pm$ 1.55% [†]	780,916 $\pm$ 96,853	39,969 $\pm$ 6,087	40.73 $\pm$ 2.43
	STGQ-Dense	1,179,859 $\pm$ 95,268	11.04% $\pm$ 1.22% [†]	773,103 $\pm$ 94,462	40,676 $\pm$ 4,788	40.70 $\pm$ 2.66
	STGQ-Routed	1,163,593 $\pm$ 91,299	11.82% $\pm$ 1.64% [†]	720,015 $\pm$ 92,196	44,358 $\pm$ 7,341	40.01 $\pm$ 2.54
	Missingness-Aware GBDT	1,320,003 $\pm$ 93,135	8.36% $\pm$ 0.97%	1,030,858 $\pm$ 114,266	28,915 $\pm$ 4,469	46.68 $\pm$ 2.45
Noise	Continuous Physical Quantile	998,273 $\pm$ 106,253	7.59% $\pm$ 0.95%	746,697 $\pm$ 82,474	25,158 $\pm$ 3,968	33.97 $\pm$ 2.78
	Frozen-Embedding Direct Quantile MLP [‡]	1,639,909 $\pm$ 242,922	0.80% $\pm$ 0.28%	1,618,439 $\pm$ 249,374	2,147 $\pm$ 953	60.62 $\pm$ 9.85
	Global Quantile	1,053,552 $\pm$ 114,150	7.48% $\pm$ 1.48%	802,361 $\pm$ 99,513	25,119 $\pm$ 5,822	36.26 $\pm$ 3.24
	STGQ-Dense	1,048,083 $\pm$ 115,181	7.63% $\pm$ 1.13%	792,472 $\pm$ 92,391	25,561 $\pm$ 4,643	36.04 $\pm$ 3.28
	STGQ-Routed	1,027,077 $\pm$ 114,317	7.62% $\pm$ 1.09%	769,864 $\pm$ 106,672	25,721 $\pm$ 3,963	35.16 $\pm$ 3.19
	Missingness-Aware GBDT	1,314,640 $\pm$ 129,158	4.66% $\pm$ 0.80%	1,160,329 $\pm$ 137,650	15,431 $\pm$ 3,099	47.11 $\pm$ 3.87
Markov	Continuous Physical Quantile	907,871 $\pm$ 96,820	7.20% $\pm$ 1.15%	663,875 $\pm$ 73,821	24,400 $\pm$ 4,106	31.59 $\pm$ 2.61
	Frozen-Embedding Direct Quantile MLP [‡]	1,680,339 $\pm$ 289,602	0.57% $\pm$ 0.25%	1,660,135 $\pm$ 298,104	2,020 $\pm$ 992	64.38 $\pm$ 12.00
	Global Quantile	1,030,131 $\pm$ 108,592	6.87% $\pm$ 1.39%	785,305 $\pm$ 97,397	24,483 $\pm$ 5,104	36.78 $\pm$ 3.35
	STGQ-Dense	1,020,078 $\pm$ 112,195	6.90% $\pm$ 1.07%	775,899 $\pm$ 96,235	24,418 $\pm$ 4,246	36.35 $\pm$ 3.57
	STGQ-Routed	990,625 $\pm$ 104,487	7.51% $\pm$ 1.22%	720,858 $\pm$ 92,754	26,977 $\pm$ 5,183	35.10 $\pm$ 3.10
	Missingness-Aware GBDT	1,261,941 $\pm$ 112,754	4.88% $\pm$ 0.75%	1,094,730 $\pm$ 120,612	16,721 $\pm$ 3,298	46.62 $\pm$ 3.40

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold numbers denote lowest cost. **Protocol Attribution:** Reports the *Clean-Calibrated Operational Protocol* at $h = 6$ (thresholds frozen on clean validation data); Continuous Physical Quantile collapses to 12.20% violation under Delay-6. For the *State-Conditional Validation-Calibrated Protocol* at immediate horizon ($h = 1$), see Section V-A2, where clean-calibrated physical rules collapse to 24.0% violation under Delay-6 while state-conditional calibration mitigates violations to 10.7% (physical) and 10.6% (joint model). Test-set iso-multipliers ($\gamma_{iso, test}$) serve strictly as post-hoc diagnostics. [‡]Frozen-Embedding Direct Quantile MLP evaluates naive uncalibrated pinball regression on frozen representations, exhibiting severe tail conservatism (>1.63 M kW reserve, $<1.2\%$ violation). In contrast, the calibrated residual architecture (STGQ-Modular, Section V-A3, Table III) and calibrated modular baselines (Cascaded Frozen MLP 15.528M $\pm$ 1.599M vs STGQ-Routed 16.065M with no statistically detectable difference) resolve this pathology.

TABLE II
SEED-PAIRED DIFFERENCE IN PENALIZED RESERVE-SHORTFALL ENERGY
AGAINST STGQ-ROUTED ($h = 6$, $\Delta \text{Cost} = \text{Cost}_{\text{BASELINE}} - \text{Cost}_{\text{STGQ-ROUTED}}$,
POSITIVE INDICATES STGQ-ROUTED SAVES ENERGY/COST AT $\rho = 10$).

Regime	Baseline Model	Δ Cost (kW-h)	95% Bootstrap CI	Sig. & Operational Verdict
Clean	Global Quantile	+39,962	[+12,556, +67,367]	Supported reduction (95% CI excludes zero)
	Cont. Physical Quantile	-77,659	[-106,175, -49,143]	Physical rule lower cost (95% CI excludes zero)
	Missingness GBDT	+279,576	[+249,539, +309,613]	Supported reduction (95% CI excludes zero)
	Frozen-Embedding Direct Quantile MLP	+690,695	[+448,769, +932,621]	Tail-conservative Direct MLP
	STGQ-Dense	+32,154	[-1,148, +65,456]	Inconclusive difference (95% CI includes zero)
Delay-6	Global Quantile	+17,010	[+3,486, +30,535]	Supported reduction (95% CI excludes zero)
	Cont. Physical Quantile	-27,126	[-46,907, -7,346]	Target exceeded (12.20% viol.)
	Missingness GBDT	+156,411	[+140,004, +172,817]	Supported reduction (95% CI excludes zero)
	Frozen-Embedding Direct Quantile MLP	+518,658	[+307,212, +730,104]	Tail-conservative Direct MLP
	STGQ-Dense	+16,267	[+7,131, +25,402]	Supported reduction (95% CI excludes zero)
Noise	Global Quantile	+26,475	[+9,121, +43,828]	Supported reduction (95% CI excludes zero)
	Cont. Physical Quantile	-28,804	[-47,555, -10,053]	Physical rule lower cost (95% CI excludes zero)
	Missingness GBDT	+287,563	[+248,847, +326,279]	Supported reduction (95% CI excludes zero)
	Frozen-Embedding Direct Quantile MLP	+612,832	[+413,733, +811,931]	Tail-conservative Direct MLP
	STGQ-Dense	+21,006	[+2,419, +39,594]	Supported reduction (95% CI excludes zero)
Markov	Global Quantile	+39,506	[+15,961, +63,052]	Supported reduction (95% CI excludes zero)
	Cont. Physical Quantile	-82,754	[-111,026, -54,482]	Physical rule lower cost (95% CI excludes zero)
	Missingness GBDT	+271,316	[+245,044, +297,588]	Supported reduction (95% CI excludes zero)
	Frozen-Embedding Direct Quantile MLP	+689,714	[+448,689, +930,739]	Tail-conservative Direct MLP
	STGQ-Dense	+29,453	[-2,874, +61,779]	Inconclusive difference (95% CI includes zero)

Note: Statistical verdicts are based on empirical 95% confidence intervals computed via day-block cluster bootstrap over the 35 test days (1,000 resamples of complete diurnal blocks). Supported reduction denotes 95% CI strictly excluding zero; Inconclusive difference denotes 95% CI including zero (indicating insufficient evidence of an operational difference resulting from shared representation learning).

state awareness, inflating recalibrated costs (824,393 kW · h at $\tau = 0$ to 1,378,900 kW · h at $\tau = 60$). In contrast, learned representations (STGQ-Modular) reconstruct operating states from electromechanical transients, achieving 755,794 kW · h at $\tau = 0$ (an 8.3% saving of 68,599 kW · h over recalibrated physics) and maintaining consistent cost savings (46.7k–68.6k kW · h) across all latencies. 3. *Regime III (Tail-Risk Bottleneck under 60-min Latency)*: Under 60-min contingency latency, unadapted physical rules collapse (24.02% $\pm$ 1.69% violation,

127.6 MWh unhedged shortfall). Condition-matching recalibration restores fleet-average violation below the 10.0% target across all models (9.46% for physical rules, 9.55% for STGQ-Modular, 9.62% for STGQ-Routed, Table III). However, fleet-average compliance conceals critical tail risk: across the five seeds, every method achieves strictly a 3/5 (60%) seed compliance rate, as Seeds 201 and 203 experience residual breaches (10.2%–10.7%).

Benchmarking selective decision abstention against random rejection and uniform margin expansion (Supplementary Table A10f) reveals that heuristic uncertainty metrics fail under multi-step SCADA latency: selective abstention (9.44% violation at $c = 0.90$) is statistically indistinguishable from random rejection (9.50%, $p > 0.40$), while uniform margin expansion strictly Pareto-dominates selective refusal (8.29% vs. 8.84% violation, saving 37,823 kW · h). Stale telemetry degrades both point forecasts and uncertainty estimators, making uniform reserve expansion and physical fallbacks more dependable than heuristic refusal.

TABLE III

MULTIDIMENSIONAL OPERATIONAL PROGRESSION ACROSS TELEMETRY LATENCY $\tau \in [0, 60]$ MIN AND PITCH OBSERVABILITY ($h = 1$, 5-SEED MEAN ON WTB; NOMINAL $\tau = 0$ BASELINE 593,258 kW · h ALIGNS WITHIN 0.6% OF THE PRIMARY FROZEN 589,535 kW · h BENCHMARK). COST IN kW · h ($\rho = 10$).

τ (min)	Pitch	Clean Phys. Cost (Viol.)	Recal. Phys. Cost (Viol.)	STGQ-Mod. Cost (Viol.)	STGQ-Rout. Cost (Viol.)	Operational Regime
0	All	593,258 (7.3%)	593,258 (7.3%)	663,347 (6.1%)	663,915 (7.3%)	Regime I (Physics Opt.)
0	None	961,024 (3.2%)	824,393 (3.4%)	755,794 (5.8%)	756,257 (6.4%)	Regime II (Repr. Adv.)
10	All	753,606 (11.2%) [†]	742,211 (7.4%)	786,366 (6.7%)	789,094 (7.6%)	Regime I (Recal. Suff.)
10	None	1,008,743 (4.8%)	928,028 (6.2%)	866,753 (6.1%)	874,156 (6.6%)	Regime II (Repr. Adv.)
20	All	896,198 (14.0%) [†]	841,734 (7.1%)	875,838 (7.4%)	879,750 (8.0%)	Regime I (Recal. Suff.)
20	None	1,062,646 (6.1%)	1,010,530 (6.2%)	947,869 (6.7%)	959,934 (7.1%)	Regime II (Repr. Adv.)
30	All	1,069,266 (16.6%) [†]	939,190 (7.5%)	987,760 (7.1%)	990,091 (7.6%)	Regime I (Recal. Suff.)
30	None	1,138,774 (7.5%)	1,101,292 (6.7%)	1,054,592 (6.4%)	1,064,238 (6.7%)	Regime II (Repr. Adv.)
60	All	1,656,285 (24.0%) [†]	1,228,609 (9.5%)	1,283,855 (9.6%)	1,274,047 (9.6%)	Regime III (Tail Risk)
60	None	1,437,280 (11.8%)	1,378,900 (9.0%)	1,330,318 (8.7%)	1,331,840 (8.8%)	Regime II (Repr. Adv.)

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest compliant operational cost. Under $\tau = 60$ min, by-seed compliance is 3/5 seeds (60%) across models.

D. Operating-Boundary Recovery and Point-Forecast Price of Routing

Evaluating point forecasting across five WTB seeds, the boundary-forced router achieves an RMSE of 236.13 ± 8.41 , compared with 224.34 ± 2.23 for iTransformer and 225.74 ± 2.60 for Graph WaveNet. Train-only inverse-frequency class weighting narrows this difference to 5.59 units (229.93 ± 2.50). This architecture is **deliberately non-RMSE-optimal**: it trades a +2.5% whole-sample error margin to preserve operating-regime observability and robust reserve hedging under telemetry degradation. Indifference to operating regimes in standard deep models triggers severe tail shortfalls (78.3–82.5 MWh unhedged exposure), whereas boundary-aware representations restrict exposure to 53.9 MWh (a 31%–35% reduction) while preserving physical regime alignment (0.8716 ± 0.0418 NMI, 0.9166 ± 0.0371 ARI; 0.721 NMI under train-only supervision). The learned model more than doubles transition recall (0.417 vs. 0.196 for degraded physics; Supplementary Figure S1). Input ablation stress guards confirm that routing fidelity is free of target leakage: withholding `Patv` or `Pab_mean` preserves mean NMI at 0.881 and 0.878 , while lagging `Patv` and lagging pitch/wind yield 0.763 and 0.684 ; gate matching peaks at the transition step (0.815) and attenuates under three- and six-step lead shifts (0.361 and 0.405).

E. Withheld-Channel Electromechanical Signature Recovery

To evaluate blind-spot resilience across commercial OEM or VPP boundaries, we test whether secondary electromechanical channels infer operating states without direct pitch measurements. In this strict counterfactual probe, withholding `Wspd` and `Pab_mean` while retaining active power yields a mean NMI of 0.561 and ARI of 0.635 across five seeds on WTB (Table IV). Removing active power (`signature_core`) drops NMI to 0.367 , whereas permuted controls collapse to 4.0×10^{-6} .

TABLE IV

SIGNATURE-GATE IDENTIFIABILITY PROBE ACROSS COMMERCIAL WIND FARMS UNDER HETEROGENEOUS PITCH OBSERVABILITY. DEFINING CHANNELS ARE WITHHELD FROM THE GATE ANCHOR; ALL VALUES ARE MEAN $\pm$ SD OVER 5 SEEDS.

Probe	RMSE	NMI / ARI	Interpretation
Canonical full-anchor	229.93	0.7208 / 0.7398	Train-only clean baseline
signature_full (no Wspd/Pab)	241.84 ± 7.19	0.5613 ± 0.0199 / 0.6347 ± 0.0239	Boundary recoverable from conseq. channels
signature_core (no Wspd/Pab/Patv)	302.10 ± 9.16	0.3671 ± 0.0372 / 0.4342 ± 0.0378	Weaker non-power signature; 3/5 seeds collapsed
sig_full_shuffled	241.50 ± 10.25	$4.0e-6 \pm 2.1e-6$ / $-1.3e-5 \pm 5.3e-5$	Negative control: chance level under permuted labels
Unconstrained MoE	—	$0.014 / —$	No declared-boundary supervision
<i>Cross-farm replication, ENGIE La Haute Borne (4 turbines, 99% pitch)</i>			
LHB canonical (full anchor)	185.0 ± 1.9	0.9752 ± 0.0088 / 0.9904 ± 0.0044	Full-anchor baseline, identical protocol
LHB signature_full	185.1 ± 1.4	0.6743 ± 0.1209 / 0.7401 ± 0.1604	Boundary signature transfers; min-seed 0.499
LHB signature_core	188.0 ± 1.0	0.5750 ± 0.0423 / 0.6270 ± 0.0638	Non-power signature transfers; 0/5 collapsed
<i>Commercial UK wind plants (Penmanshiel MM82, Kelmarsh MM92)</i>			
Penmanshiel signature_full	—	0.3024 ± 0.0343	Above the 0.20 criterion; shuffled 4.4e-5
Penmanshiel signature_core	—	0.1954 ± 0.0961	Borderline: below 0.20 but 4400x above shuffled chance
Kelmarsh signature_full	—	0.3402 ± 0.0736	Above the 0.20 criterion; shuffled 9.2e-5
Kelmarsh signature_core	—	0.3775 ± 0.0783	Above the 0.20 criterion

Cross-farm probes confirm that electromechanical consequence channels recover operating regimes across turbine technologies without direct pitch telemetry (Section VI).

F. Degradation Resilience and Early Warning Dynamics

Under fair input degradation (Supplementary Table A10d), clean deterministic thresholds exhibit high sensitivity (recall 1.000 vs. 0.9705; logistic regression achieves 1.000 recall with 0.879 precision vs. 0.630 for the routed posterior). Under 6-step latency ($d = 6$), however, deterministic rule recall collapses from 1.000 to 0.196 (0.249 F1). In contrast, the boundary posterior sustains an early-window recall of 0.4170 ± 0.1137 (+0.221 gain) with F1 of 0.3802. When supervisory labels drop to 50%, the learned model sustains 0.971 recall against 0.508 for threshold rules (+0.462 gain). Under Markov-Gilbert burst dropouts ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$), stale thresholds drop to 0.840 recall, whereas the routed posterior sustains 0.962 ± 0.023 recall and 0.925 ± 0.042 F1 (seed-paired F1 gain +0.021, 95% bootstrap CI $[-0.014, +0.050]$), confirming dynamic wake graph defense-in-depth.

G. Boundary-Window Reserve Vignette and Benchmark Comparison

At shortage penalty ratio $\rho = 10$ on the transition boundary band, the boundary-router gate-bin policy costs 95.13M, outperforming the same-model global rule (99.55M), while low-RMSE Graph WaveNet with physical-bin reference achieves 84.31M (≈ 1.28 M kW·h per single dispatch interval), avoiding ~ 637 MWh-equivalent shortage cells.

H. Point of Common Coupling (PCC) Portfolio Smoothing & Longitudinal Drift

Aggregating decentralized turbine power flows at the Point of Common Coupling (PCC) bus ($P_{\text{farm}} = \sum_{i \in \mathcal{V}_t} P_{i,t}$, ~ 121 active turbines) unlocks spatial portfolio diversification: high-frequency turbulence cancels across the array, leaving boundary transitions as the primary reserve risk. Across unstratified full-year envelopes, paired differences between individual summing and farm-level aggregation are statistically inconclusive (mean difference -2.32 M kWh, 95% CI $[-6.07\text{M}, 1.73\text{M}]$ vs. global PCC quantiles; -1.37 M kWh, CI $[-6.33\text{M}, 2.82\text{M}]$ vs. Gaussian sizing). Crucially, significant diversification concentrates in transitional regimes (10%–90% pitching), saving -1.33 M kWh (CI $[-1.68\text{M}, -1.07\text{M}]$) over continuous physical pitch. Longitudinal records across Kelmarsh (9 years) and Penmanshiel (8.6 years) illustrate periodic schedule recalibration under climatological drift.

VI. DISCUSSION

A. Operational Diagnostics and Deployment Boundaries

The boundary-aware architecture functions as a targeted diagnostic for pre-dispatch reserve risk screening, prioritizing transition-window tail reliability over whole-sample point accuracy. Pristine SCADA telemetry favors unconstrained architectures (Graph WaveNet, iTransformer); conversely, delayed or pitch-withheld telemetry requires boundary-risk posteriors for tail defense. Industrial deployment is governed by three empirical axes: 1. *Directional transfer boundaries*: Zero-shot cross-farm transfer exhibits pronounced asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.752–0.770, ARI 0.735–0.761 vs. Penmanshiel $\rightarrow$ Kelmarsh NMI 0.341–0.505, pooled mean NMI 0.557). Unadapted transfer is bounded by turbine rotor dimensions and array aerodynamics; reliable transfer requires local recalibration. 2. *Within-plant local retraining*: Chronological

retraining across ENGIE La Haute Borne (five-seed routing NMI 0.941, ARI 0.971; canonical NMI 0.975; withheld probe 0.674/0.575, collapsing to 0.001 without boundary anchors), Kelmarsh (probe 0.340/0.378), and Penmanshiel (probe 0.302, signature_core 0.195) confirms neural architectural adaptability to site-specific turbine technologies. 3. *Longitudinal rolling evaluation*: Across decadal archives (Kelmarsh, 9 years; Penmanshiel, 8.6 years), sliding 2-year window evaluations demonstrate operational recalibration under decadal climatological drift.

On compact micro-farms like ENGIE La Haute Borne (4 turbines, rolling terrain), spatial wake propagation offers limited physical redundancy, causing spatial graph convolutions to overfit localized terrain features during seasonal shifts (+1.01M kWh cost delta vs. global quantile, 95% CI [+0.29M, +1.76M]; Supplementary Table A11h). Expanding calibration to 180 days compresses this gap to +43k kWh (CI [-28.7k, +141.2k]). Complex spatio-temporal architectures are unnecessary on micro-farms under clean telemetry, and should be reserved for utility arrays or degraded/pitch-sparse conditions. Downstream flexibility coordination (such as battery energy storage) represents a natural operational extension (illustrated in Supplementary Material).

VII. LIMITATIONS

WTB pseudo-labels derive from anemometer wind speed and pitch angle. Counterfactual withheld-channel tests confirm boundary recovery via consequence signatures across four commercial plants (permuted controls at chance). Operational regime identification functions as an anchor-constrained diagnostic grounded in engineering control boundaries rather than unsupervised physical law discovery [27]. The analytical scope is formally bounded to Level 1 local pre-dispatch risk screening proxies (PSREI at $\rho = 10$), abstracting away Level 2 transmission network clearing with Locational Marginal Prices, full AC-OPF power-flow constraints, and multi-settlement cashflows [24], [34].

VIII. CONCLUSION

This study establishes an empirical operational boundary and value-of-information framework delineating aerodynamic power rules, state-conditional recalibration, and learned representations under SCADA telemetry degradation. Across 5 seeds on the 134-turbine WTB plant, continuous aerodynamic quantile rules minimize reserve-screening surrogate cost under pristine telemetry (589,535 kW · h at $h = 1$, 881,367 kW · h at $h = 6$), but suffer systematic reliability breakdown under a 60-min latency contingency, surging violation rates to $24.0\% \pm 1.7\%$ ($h = 1$) and $12.20\% \pm 1.25\%$ ($h = 6$).

Under full channel observability, state-conditional recalibration recovers violation to 10.7% and cuts shortage from 127.6 MWh to 57.0 MWh (55.3% reduction) at lower cost than neural models. When pitch registers are unobservable across aggregator boundaries, learned representations reconstruct operating regimes from electromechanical transients, outperforming recalibrated physics by 46.7k–68.6k kW · h across all latencies and doubling regime recall (0.416 vs. 0.196). Dynamic Mixture-of-Experts routing achieves sta-

tistical parity with unrouted dense baselines (cost ratio 1.045, $p = 0.380$), while decoupled modular architectures match tail reliability (9.7% violation), indicating that operational value stems from shared spatio-temporal representations and residual calibration rather than dynamic routing. Reliable cross-farm deployment requires local retraining (ENGIE La Haute Borne, NMI = 0.941). These findings position machine learning as an accountable diagnostic instrument for resilient wind plant operations, operating alongside calibrated physical curves under verifiable communication boundaries.

IX. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

X. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmarsh, and Penmanshiel SCADA datasets are public (raw third-party data not redistributed). Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WTB routing, reserve audit, anchor-stress caches, early-warning label degradation, classifier control, class-weight sensitivity, external diagnostics, and evidence-freeze protocols across declared seeds. Results are statistically reproducible across declared seeds rather than bitwise deterministic across all GPU/CUDA environments. The analysis involves no human subjects.

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